

Link:

<https://hailtokiyohime.github.io/3D-Nightingale-s-Coxcomb/>

Design Overview

In this assignment, I chose to redesign Nightingale's *Coxcomb* as an interactive 3D visualization built in Unity. Instead of presenting the graphic as a static historical diagram, I turn it into a 3D object that allow the user to explore the data interactively.

This redesign keeps the original core idea of representing causes of mortality over time, while providing a new way for users to compare the proportion different of each data more intuitively.

Key Modifications

The project was developed in Unity and exported into Github. When opened, the viewer is presented with a 3D version of Nightingale's coxcomb, changing it into an exploratory data.

The graph is interactive. Users can spin the graph with the mouse wheel, rotate it by dragging with the left mouse button, move the camera by dragging with the right mouse button, and zoom in or out by holding Ctrl while using the mouse wheel. These controls were added to let viewers examine the form from multiple angles and better understand the relative scale of different sections.

When the user mouse hover on top of a section, it will highlight the section, and display the detail of that month.

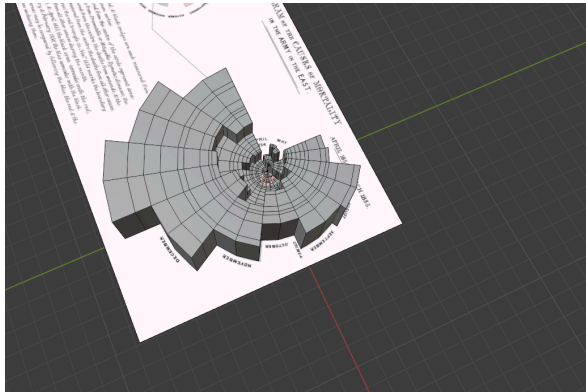
Rationale

When Nightingale originally created the coxcomb chart, she intentionally used the area of each sector to represent the number of deaths. This approach creates a strong visual impact and gives the chart a high degree of persuasive power. However, although area can be visually striking, the human eye is not very good at precisely comparing differences in area. As a result, viewers may find it difficult to understand the proportional relationships between months.

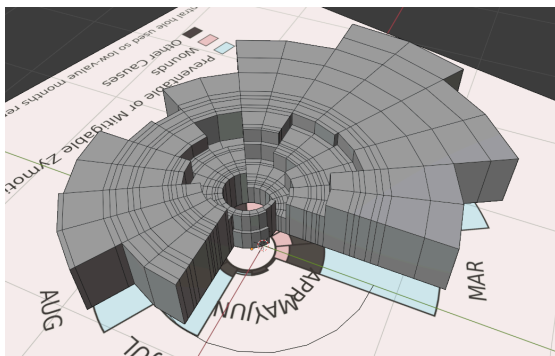
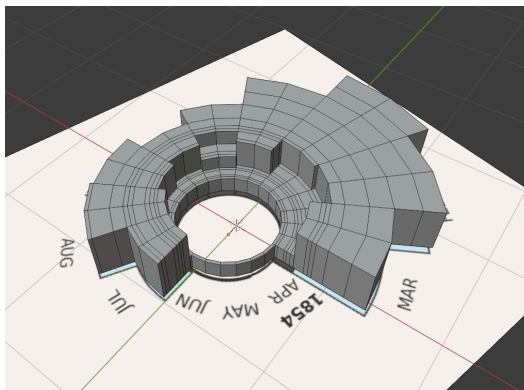
In my redesign, I introduced height as an additional dimension. This allows each section to preserve the original radial structure when looking from the top while also using vertical height to further represent the magnitude of the data when looking at a 45 degree angle. This makes it easier for viewers to understand the proportional gaps in the data and to identify which months had especially high or especially low mortality.

Iteration:

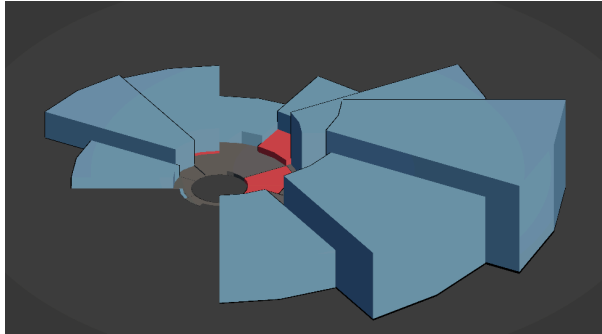
I use the original Nightingale's Coxcomb as the draft and create a rough blender model with it. However, as you can see in this image, the center became visually crowded because the inner sections were compressed together, especially when each section also had a different height.



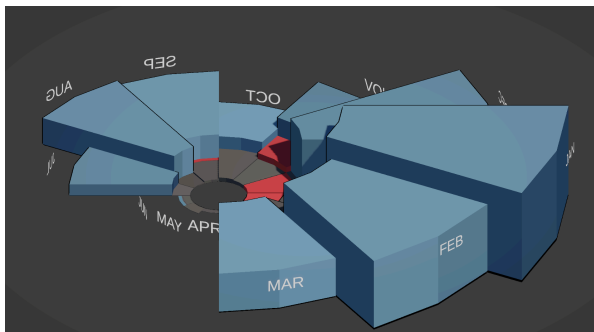
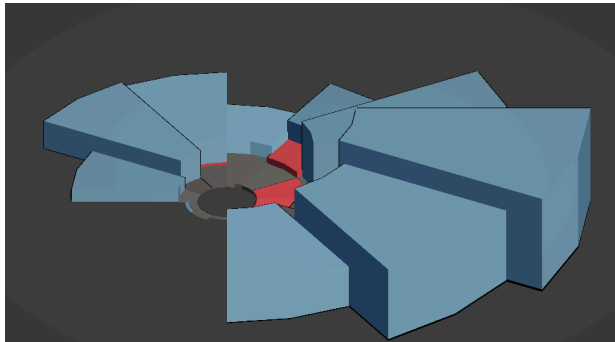
So I introduced a hole in the center of the graph. I recalculate the graph and build the graph again. This added visual space, reduced clutter, and improved clarity.



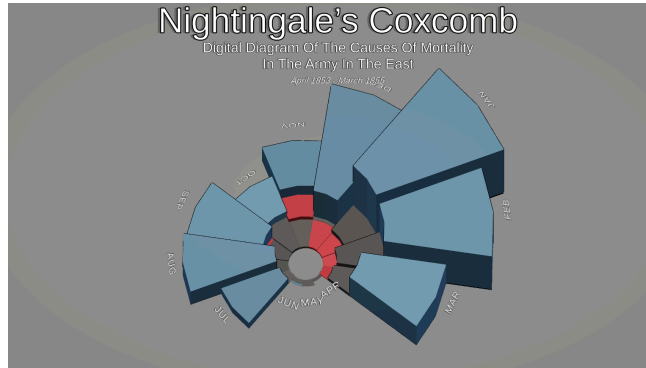
Then I improved the graph into Unity. Give each section a different color and raise the height based on the number of deaths.



I added slight color variations between sections that represent the same cause of death, so that even when they use the same base color, the sections do not visually blend together.



Final Product:



One cool aspect of this graph is, when viewed from above, it appears as a traditional Nightingale's coxcomb with a hole in the center.



Looking at a 45 degree view allows human eyes to precisely comparing differences in height.

